

Generative AI in the secondary classroom

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Session overview

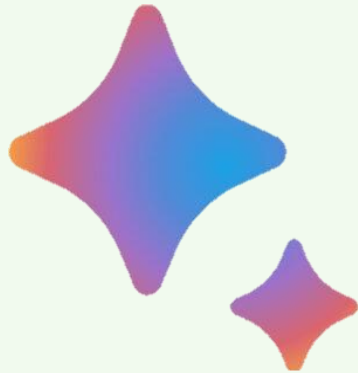
- To discuss how the use of AI can support teacher workload
- To understand how your students may be using AI
- To consider how to teach about AI

What is your understanding of **Generative AI?**

How do you use them?

How do they work?

How might teachers and children interact with these tools?



AI



Automation of knowledge work

Artificial Intelligence

Narrow AI

Data-driven AI

Machine Learning

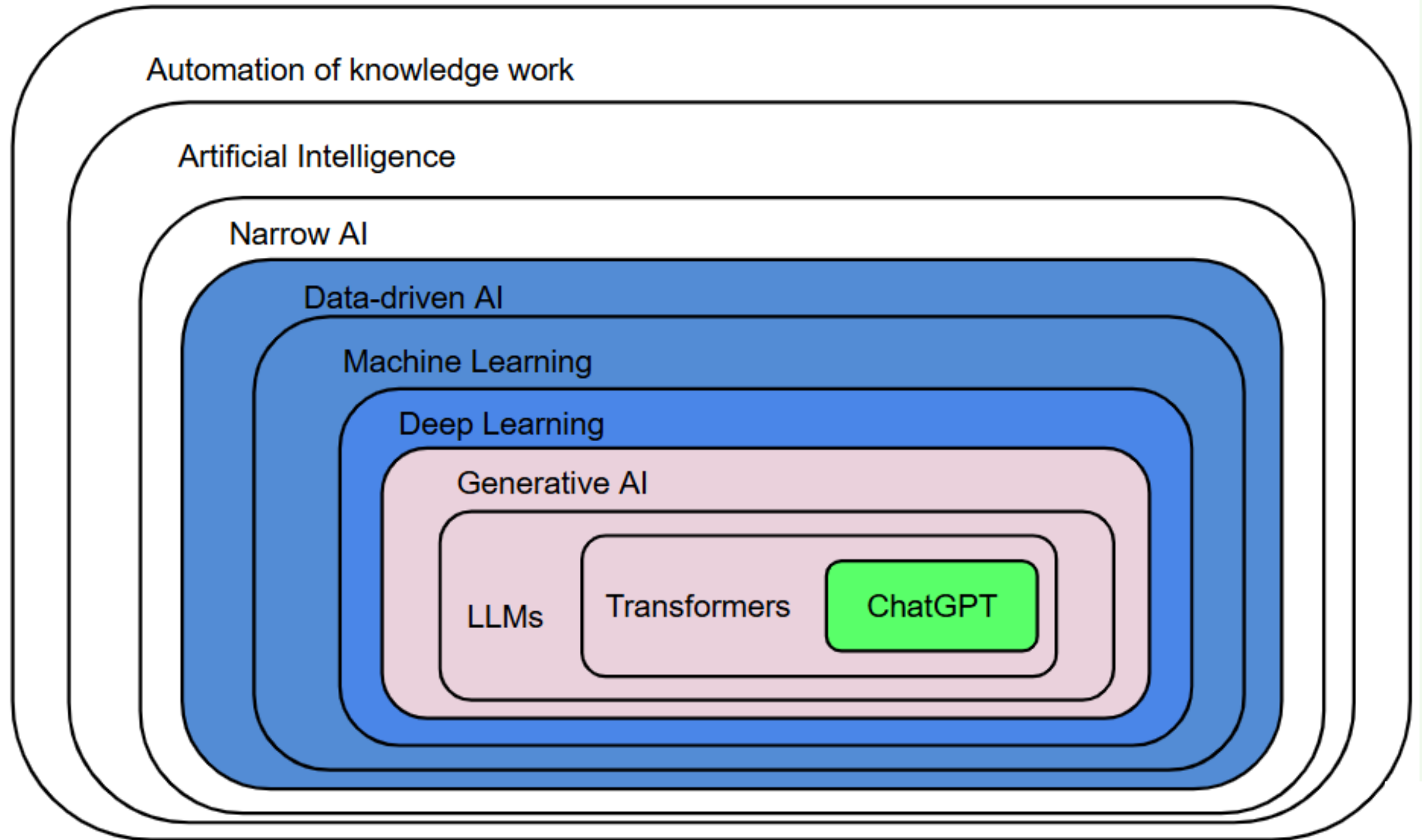
Deep Learning

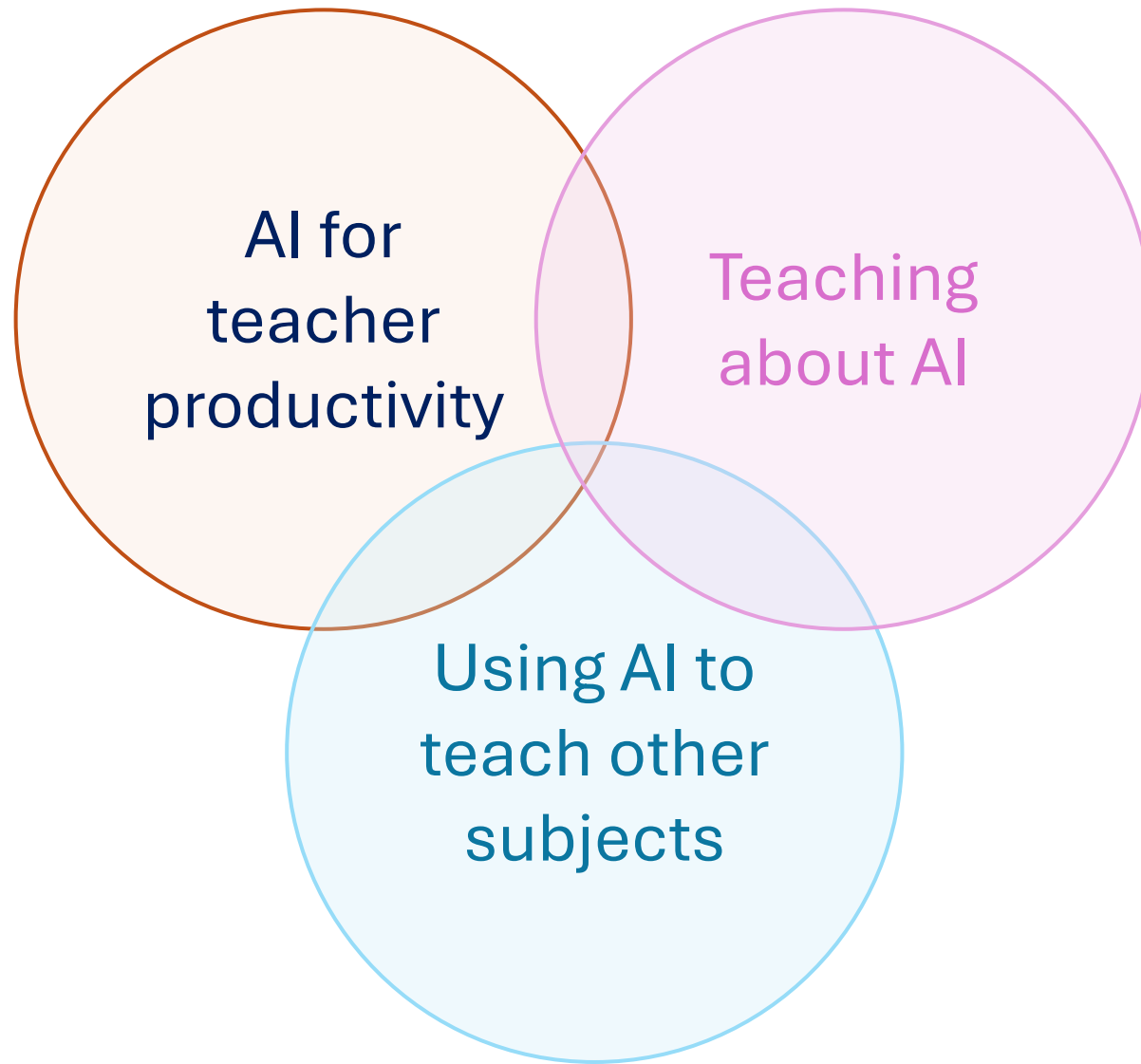
Generative AI

LLMs

Transformers

ChatGPT





How the others are talking about AI in education

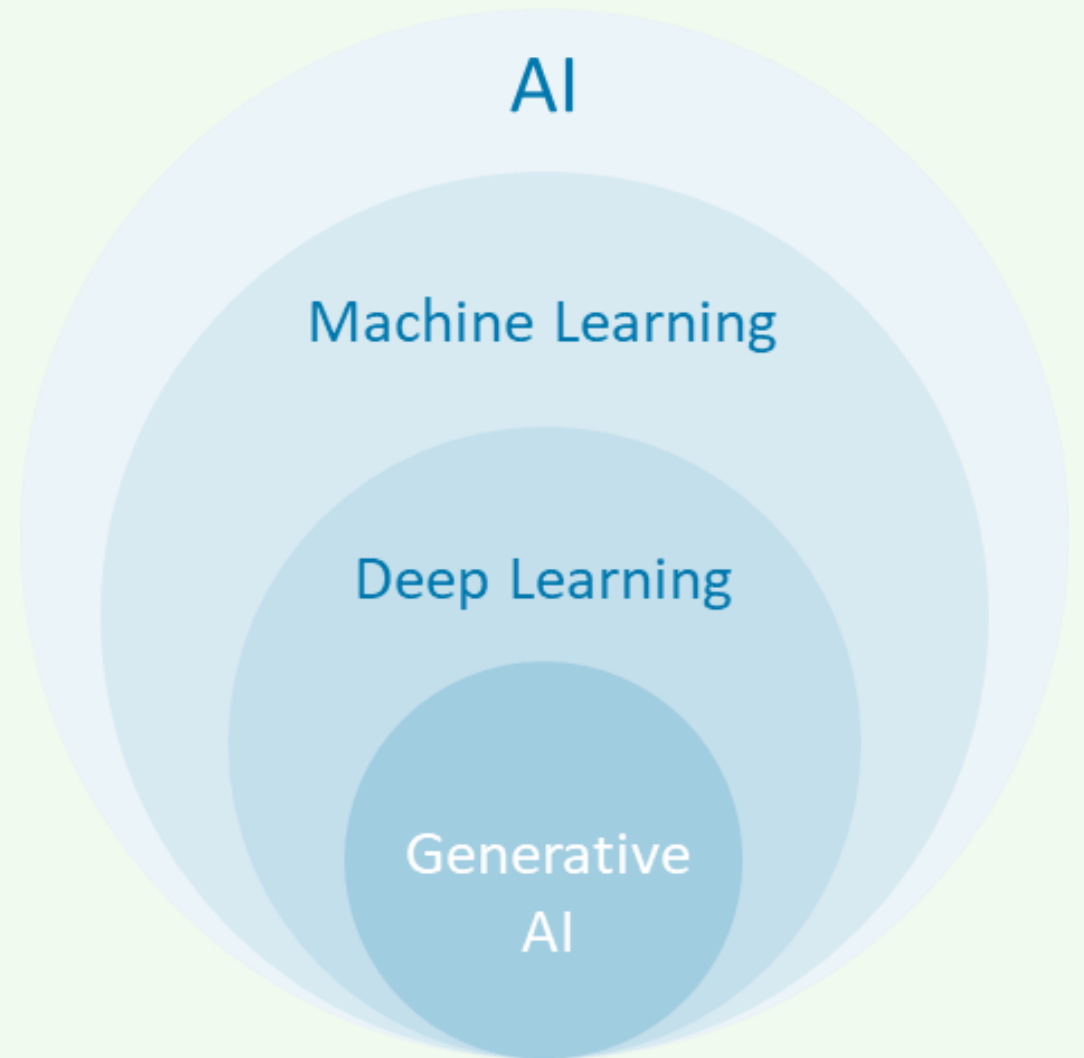


Raspberry Pi Foundation (2024)

AI vs GenAI

Generative AI tools, which we are defining as tools that generate new content, such as images or text, to a response given by a user.

(Fengchun and Holmes, 2023)



Learning about Generative AI

How these tools are being used in schools by teachers and their students.

Learning about what to expect while on your placements and how they you can utilise these tools to support teaching and learning.

Using generative AI tools to support your academic development.

Learning about how to use these tools ethically and professionally to support your studying.

How do these tools work?



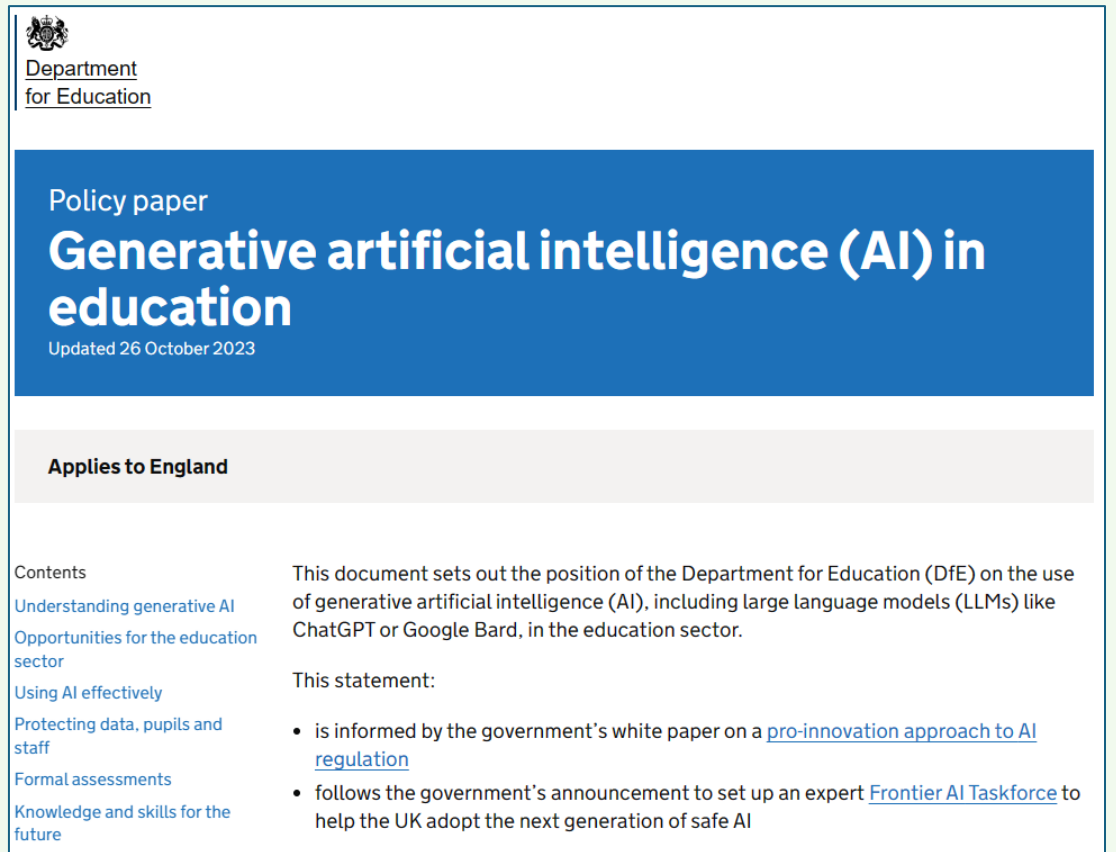
Further explanation

- **Corpus** of data is gathered.
- Models are trained on this corpus.
- Users enter a **prompt** to the tool.
- The tools pulls information from the corpus.
- It presents an answer in the form of a cohesive answer.
- Warning: they can **hallucinate** and put your prompts back into the corpus.

Reflection: what skills do we have as humans, that cannot be replicated by these tools?

What does the DfE say?

- ***Opportunities & Challenges:*** Offers potential benefits for education but also presents challenges that require careful management.
- ***Professional Responsibility:*** Emphasizes that the final quality of documents remains the responsibility of the individual and their organization².
- ***Data Privacy & IP Rights:*** Highlights the importance of complying with data protection laws and respecting intellectual property rights of student.



The screenshot shows the cover and introductory text of a DfE policy paper. At the top left is the DfE logo and name. The main title 'Generative artificial intelligence (AI) in education' is in large white text on a blue background, with 'Policy paper' above it and 'Updated 26 October 2023' below it. A grey box below the title states 'Applies to England'. On the left, a 'Contents' section lists: 'Understanding generative AI', 'Opportunities for the education sector', 'Using AI effectively', 'Protecting data, pupils and staff', 'Formal assessments', and 'Knowledge and skills for the future'. The main text area on the right states the document's purpose and lists two key points under 'This statement:'.

Department for Education

Policy paper
Generative artificial intelligence (AI) in education
Updated 26 October 2023

Applies to England

Contents

- Understanding generative AI
- Opportunities for the education sector
- Using AI effectively
- Protecting data, pupils and staff
- Formal assessments
- Knowledge and skills for the future

This document sets out the position of the Department for Education (DfE) on the use of generative artificial intelligence (AI), including large language models (LLMs) like ChatGPT or Google Bard, in the education sector.

This statement:

- is informed by the government's white paper on a [pro-innovation approach to AI regulation](#)
- follows the government's announcement to set up an expert [Frontier AI Taskforce](#) to help the UK adopt the next generation of safe AI

DfE (2025)



Let's have a look at one of these generative AI tools.

Copilot is available through our Microsoft accounts.

Microsoft Bing SEARCH COPILOT NOTEBOOK 21666@stma... Protected

Copilot

Your everyday AI companion

Laugh

What's a good joke to start my 6th-grade math class?

Analyze

How can teachers make lectures more engaging and effective?

Summarize

Tell me the latest findings of the Nation's Report Card in five paragraphs

Create

Help me create guidelines for 7th graders on writing a persuasive essay

Write

Write a grant proposal to take 10th graders to a college fair

Templatize

Write a template for a 5th-grade reading log

Copilot uses AI. Check for mistakes. [Terms](#) | [Privacy](#) | [FAQs](#) | [Learn more](#)

Choose a conversation style

More Creative

More Balanced

More Precise

Your personal and company data are protected

Ask me anything...

0/2000

Your turn!

Task

Using a lesson you have recently taught, ask the tool to suggest activities to teach that LO.

Reviewing the output from the tool ...

- Can the ideas realistically be delivered in one lesson?
- Does the lesson consider all learners?
- Is the activity pitched too high?
- Does the idea include everything needs to make a great lesson?
- Is there challenge evident?

Prompt engineering

Give it a role: Tell the chatbot to adopt a role such as a subject expert or a student.

Task: Clearly define the task or goal for the chatbot prompt.

Steps: Outline the specific steps that the chatbot should take to complete the task.

Context: Provide context for the task, such as the reading age of the audience.

Goal: Specify the desired outcome for the chatbot prompt. Provide an example framework or headings as a template.



Planning and resourcing lessons

Task 1

Create age-appropriate definitions of key vocabulary for your lesson.

Task 2

Ask the tool to generate a story to make some learning memorable for the children.

Task 3

Plan a multiple-choice quiz for a lesson you plan to teach. Ask the tool to anticipate and include misconceptions.

Task 2

You need to email a parent about some challenging behaviour, get the tool to produce a template.

Task 3

You want to find a list of jobs for children to potential do if they enjoyed a specific unit of learning in Science.

Reflective practice examples

Task 1

Find out some standard experiences the children may have in learning or life for the new unit of learning.

Task 2

Tell the tool about your lesson and ask it to anticipate things that may happen or things you have forgotten.

Task 3

Ask the tools to generate a scenario for supporting a distressed pupil. Interact with the scenario.

What do we know from the classroom?



St Mary's
University
Twickenham
London
School of
Education

Perceptions of Generative Artificial Intelligence from the classroom

**Student and staff views and
experiences, and implications
future practice**

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Primary students

Held misconceptions about the power of these tools, especially relating to hacking.

Limited use to support their learning and away from their learning.

There was an interest to find out more and be taught about some of these tools.

Secondary students

Students were using these tools more frequently away from learning.

Progression in the sophisticated use of these tools.

They discussed how they use the tools now with limited discussion of their use in future jobs.

Teachers

Are unsure that these tools can help with the parts of the job they'd like the most help with.

See the tool as an addition to their practice.

Want further support with how these tools can be utilised by them and their students.

Key findings from our research

Using the tool as a more efficient Google search as it presents the information cohesively.

Seeking advice from the tool about things that are going on in their lives.

Using these tools to ask questions and check their understanding of elements of their learning.

More creatively, these tools were used to create new content such as unique images and new recipes.

Specific examples – students

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Key findings from our research

Removes the blank page starting point for planning and resource generation.

Generating a story to support the learning with specific characters or events that will interest the children.

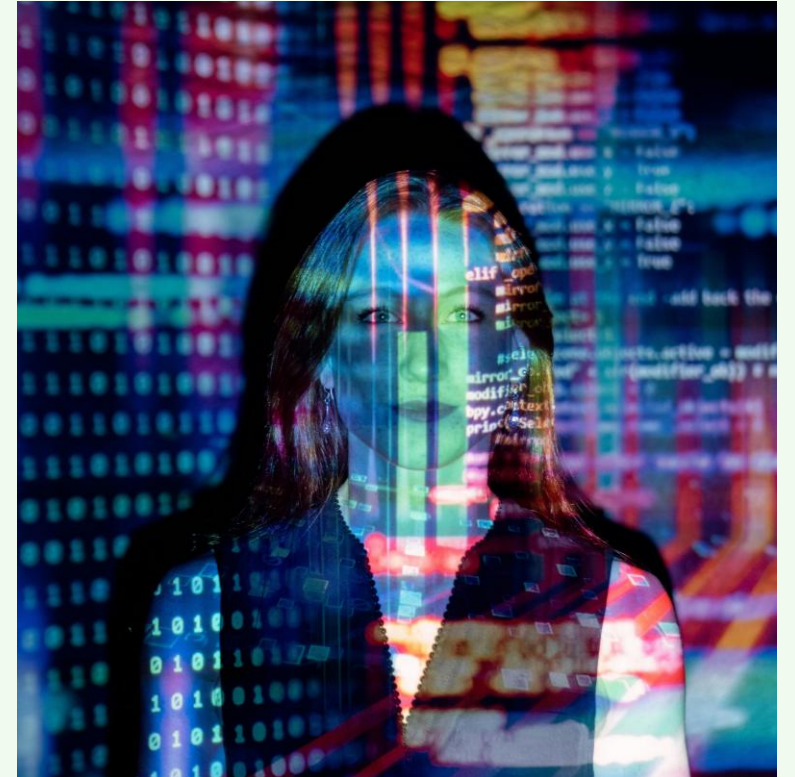
Using the tools to summaries texts, such as DfE policies so that they do not have to read the entire thing.

Using the tool to generate a lab report and having this as a starting point for a discussion with children.

Specific examples – teachers

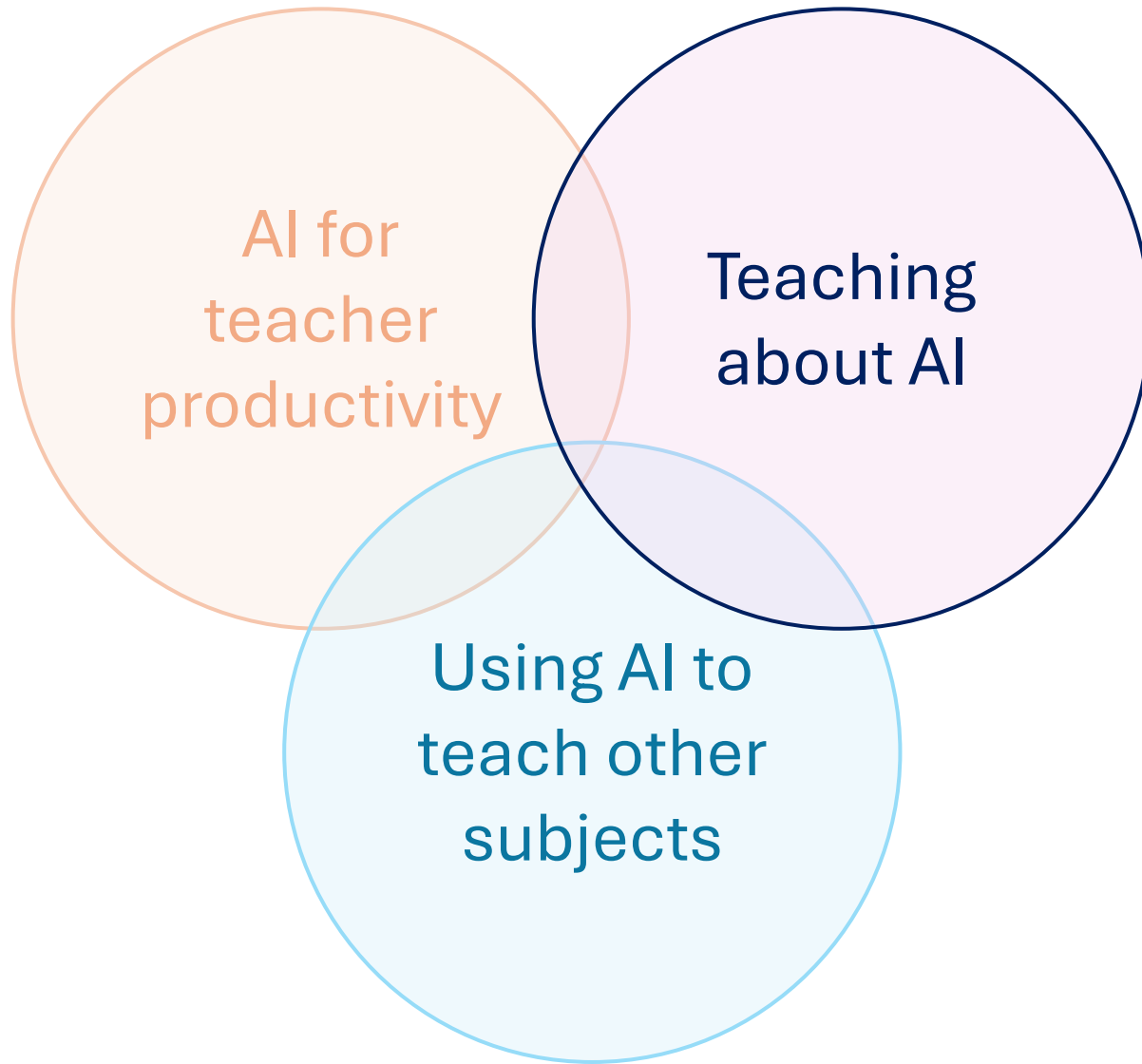
Themes across all groups

- Positives and challenges
- Application of generative AI for learning
- The need for further learning



Key considerations for our schools

1. To invest in *staff knowledge and understanding* in the field of generative artificial intelligence, so they feel equipped to teach and discuss this with their students.
2. To *consider the place of generative artificial intelligence in the curriculum* so that students are taught how to ethically utilise the tools and consider their roll in future industry.
3. The importance of promoting *consistency in schools in discussing generative artificial intelligence amongst staff*, to ensure key messages are communicated to students.



Why do you think
we should be
teaching young
children about
artificial
intelligence?

Welcome to my digital talent show



Welcome to my digital talent show

She was a stern-looking witch with square glasses perched on her nose and black hair usually pulled back into a tight bun. She wore an emerald-green robe, reflecting her position as a professor.

A tall, black-haired witch in emerald-green robes stood there. She had a very stern face and Harry's first thought was that this was not someone to cross.

What if these tools were used to create news?

The Heroic Cat

A cat in London saved its family from a fire! The cat noticed the smoke and ran to wake up its owners by meowing loudly and scratching at their bedroom door. Thanks to the cat's quick actions, the family was able to escape safely.



The future of your classrooms

In a futuristic city, schools have started using robot teachers! These robots can teach all subjects and even help students with their homework. The students love their new robot teachers and are learning faster than ever before.



Device ownership & internet use

- 9 in 10 children will have their own smart phone by 11. [\(Woodhouse and Lalic, 2024\)](#)
- Almost all children (96%) between the ages of 3 and 17 go online regularly via range of smartphones, tablets and laptops. [\(Ofcom, 2024a\)](#)
- Half of children aged 3-12 use at least one social media app/site despite the minimum age requirement of 13. [\(Ofcom, 2024a\)](#)
- A third of 8-17-year-old children are being exposed to content online that they find worrying or nasty. [\(Ofcom, 2024a\)](#)
- Children are most likely to see harmful content that uses offensive language (35%), shares misinformation (33%) or receive an unwanted follow request (27%). [\(Ofcom, 2024b\)](#)

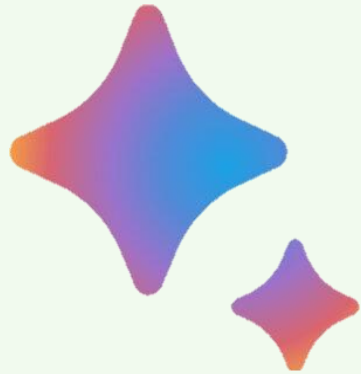
‘Children grow up with technology, own their own devices and regularly use the internet – they know how to use it effectively!’

Challenging the ‘digital native’ mindset

- ‘Exposure to technology does not equate with ability to use it.’ (ICDL, 2014)
- Despite often being regular users of technology, they may lack basic digital literacy skills that they need to engage with digital technology to support learning. (Muller and Goldenberg, 2021)
- ‘One of the barriers to pupils developing knowledge in digital literacy is the belief that they are ‘digital natives’ and already experts in the use of digital devices.’ (Ofsted, 2022)

‘Defining what is meant by digital literacy however has proven complicated, as the spaces, texts and tools which contextualise such practices are continually changing.’

(Pangrazio, 2016, p163)



AI



A critical approach to digital literacy should be taught where the interaction of power and identity should be considered.

(Pangrazio, 2016)

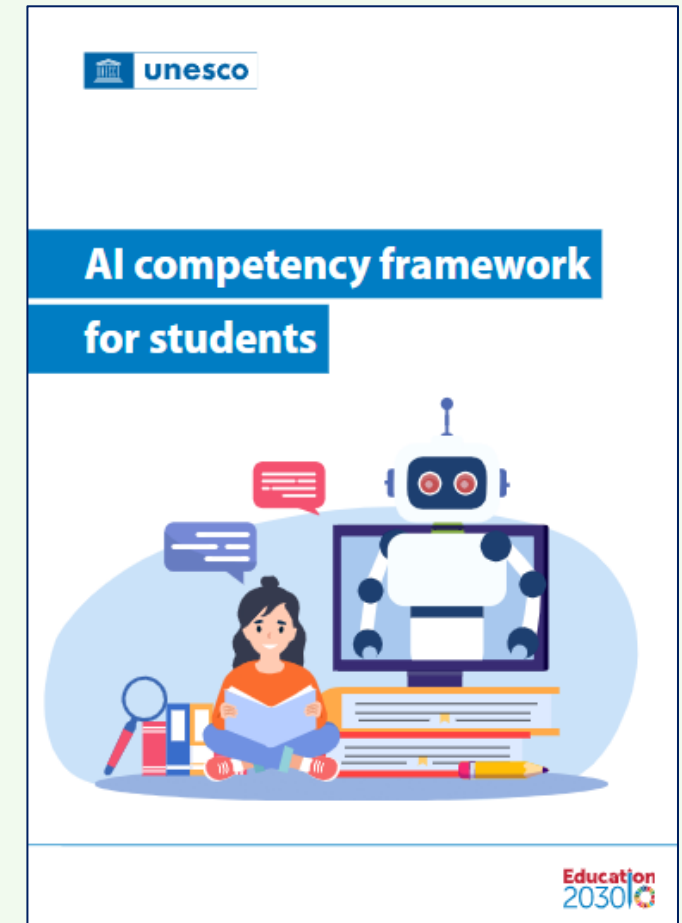
What can educators do?

- Challenge those educator digital native thoughts.
- If using technology in front of the students discuss how you're using it and model effectively.
- Talk to the students about their uses of technology.
- Maximise all opportunities to teach digital literacy, the CoE framework is a good starting point.
- Consider the implications of power when talking about digital content with students.

UNESCO AI competency framework

Competency aspects	Progression levels		
	Understand	Apply	Create
• Human-centred mindset	• Human agency	• Human accountability	• Citizenship in the era of AI
• Ethics of AI	• Embodied ethics	• Safe and responsible use	• Ethics by design
• AI techniques and applications	• AI foundations	• Application skills	• Creating AI tools
• AI system design	• Problem scoping	• Architecture design	• Iteration and feedback loops

The UNESCO framework teaches children about AI, but to also be critical and ethical thinkers about AI



Further resources

Practical resources:

[TeachMateAI](#)

[Magic School AI \(USA focus\)](#)

[Tool for children to see how AI works – drawing](#)

[Tool for children to see how AI works - animation](#)

[KS3 Raspberry Pi resources](#)

Further reading:

[AI in education group](#)

[AI for Inclusive Learning](#)

[HelloWorld magazine on AI in education.](#)

[RaspberryPi blog](#)

Wider conversations about AI in education

The need for ***expert knowledge*** when using Generative AI tools to ensure you know when content is inaccurate.

How should children be taught about these tools and this ***influence on their lives.***

The ***role of knowledge within education*** and whether tools that mean knowledge is accessible easily will change education.

The ***wider impact of these tools on the environment*** and the influence they may have on sustainability goals.

Continue the discussion ...

Teacher workload

What should I be cautious of?
How effective is it for my subject?
How can I ensure I remain
critical?
Where can I seek advice on the
content it produces?
Human in the loop.

Student's use

Are they being taught how to
ethically use this?
Do wider discussions about
ethics and sustainability need to
be had?
Is it a support or hinderance?
Is there a place for these tools
with learning?
Safeguarding!

Generative AI key vocabulary

Artificial Intelligence	Is a wide-ranging branch of computer science concerned with building smart machines capable of performing tasks that typically require human intelligence.
Large Language Models (LLM)	A form of AI that is trained on a mass of data. They can communicate and generate human-like text in response to a wide range of prompts and questions.
Generative Artificial Intelligence	A subset of AI that is concerned with generating new content.
Corpus	The large sets of data that LLM have been trained on. This data set varies from tool to tool.
Unconscious bias	The hidden or automatic judgments and opinions we form about people, places, or things without being consciously aware of it
Prompt engineering	The term coined for the art of giving the tools a prompt to produce a response. The better the prompt, the better the answer.
Hallucinations	When a tool gives a response that includes false information. Tools often do this when they do not have the information in their corpse .